DELIVERABLE: ONE CONSOLIDATED POD REPORT

1. Dataset overview: purpose, structure, number of records and attributes

Purpose: Data for building the Sentinel LM.

Structure and number of records: 14 months of labelled transactions (about 1.2 crore rows, Jan 2025 – Feb 2026)

Attributes

Field	Type
transaction_id	string
customer_id	string (hashed)
transaction_amount	decimal (₹)
merchant_category	categorical
transaction_time	timestamp (IST)
device_id	string (hashed)
customer_age	integer (years)
customer_location	lat/long + city
merchant_location	lat/long + city
previous_transactions	integer
average_transaction_value	decimal (₹)
failed_transactions	integer
account_age	integer (days)
fraud_label	binary (0/1)

2. Data dictionary: every attribute with its data type, business meaning and example values

Field	Type	Description	Example
transaction_id	string	Unique transaction reference	TX98231
customer_id	string (hashed)	Pseudonymised payer identifier	C_7f31a9
transaction_amount	decimal (₹)	Value of the transaction	50000.00
merchant_category	categorical	MCC group (travel, gaming, electronics, etc.)	electronics
transaction_time	timestamp (IST)	Date and time of the authorisation request	2025-11-14 03:12:44
device_id	string (hashed)	Fingerprint of the device used	D_44be20
customer_age	integer (years)	Payer's age from KYC	27
customer_location	lat/long + city	Location of the payer at transaction time	Pune
merchant_location	lat/long + city	Registered location of the merchant	Delhi

Field	Type	Description	Example
previous_transactions	integer	Count of the payer's transactions in the last 30 days	14
average_transaction_value	decimal (₹)	Payer's mean transaction value over 90 days	2,350.00
failed_transactions	integer	Failed attempts by the payer in the last 24 hours	4
account_age	integer (days)	Days since the payer's account was created	6
fraud_label	binary (0/1)	1 = confirmed fraud, 0 = genuine	1

3. Business mapping: how the dataset relates to the SENTINEL client requirements

The main goal of the Sentinel LM project is to prevent or reduce fraud. In the current dataset we have attributes like transaction amount, customer_location, merchant location, account age, etc. The example rule that was provided to us in the Project document was:

> *"A ₹50,000 transaction is not suspicious by itself. The same ₹50,000 from a new device, in a new city, at 3 AM, right after several failed attempts, very likely is. Fraud is a combination of context, behaviour and timing, which rules capture poorly and which a learned model can capture well."*

So the data basically has all these data points and will be the main source for building the model.

4. Data quality observations: missing values, duplicates, inconsistencies and other findings

The file is clean structurally: it parses fully, every ID is well formed, and there are no duplicates. The problems are in a few missing fields, some extreme amounts, column names that don't match what the data holds, and how the train/validation/holdout splits are built.

4.1 Missing values

Only four columns have gaps. All of them are empty strings rather than NA/null text.

Column	Missing	Pattern
customer_location	3,257 (3.0%)	Spread across all channels and splits; only 62 fall on a customer's first transaction
average_transaction_value	2,829 (2.6%)	Not random. It's empty exactly when the customer had no transaction in the previous 90 days (2,480 are first transactions, the rest follow gaps of more than 90 days)
device_id	1,138 (1.0%)	All UPI. 858 of them are in the train split
everything else		—

4.2 Duplicates

- No full-row duplicates, no repeated transaction_id, and no duplicates even when the ID is ignored.

- There are 318 back-to-back transactions from the same customer less than 1 second apart (smallest gap: 2 ms), and 3,169 less than 10 seconds apart. These could be bursts of fraud, or retries or double submissions that need deduplicating.

4.3 Inconsistencies and misleading semantics

- **Redundant columns:** amount_paise always equals transaction_amount × 100, with 0 mismatches.
- **Constant column:** is_synthetic is true on every row, so it carries no information.
- **Timestamp precision varies:** milliseconds have trailing zeros stripped (.602, .73, .9, or none at all). Pandas parses them fine, but strict parsers may not. Can be removed.
- **Customer location jumps around.** 2,080 of 2,480 customers appear in 2 to 9 different cities, with 9,002 city changes in total. 20 of those changes happen within an hour (4 within 10 minutes), which is physically impossible.
- **Shared devices:** 120 devices are used by more than one customer, one of them by 17. 2,141 customers use more than one device (up to 8).
- **Age vs account age:** on 4,942 rows the account is older than the customer's adult years (account_age/365 > age − 18). That includes 1,986 rows where an 18–19-year-old has an account more than 2 years old.
- **account_age = 0 on 25 rows.** 24 customers have a constant account_age; most of them have only 1–3 transactions.
- What *is* consistent: each customer has one age, each merchant has one category and one location, each city name always has the same coordinates, account_age grows exactly with elapsed time, and failed_transactions never exceeds previous_transactions.

4.4 Outliers

- **Very skewed amounts:** the median is ₹1,250, the 99th percentile ₹25k, and the maximum **₹79.96 lakh**. 147 transactions are over ₹1 lakh and 17 are over ₹50 lakh, typically about 37× the customer's own average.
- **Implausible for the channel:** 97 UPI transactions are over ₹1 lakh and 28 are over ₹5 lakh (UPI's per-transaction caps are around ₹1–5 lakh). Some are in categories like grocery and quick-commerce. They look like injected anomalies.
- These outliers inflate the rolling average: 413 rows have average_transaction_value over ₹1 lakh.
- 4 transactions are under ₹10 (the minimum is ₹8.20).

4.5 Splits and coverage

- The splits are cleanly time-based on IST month boundaries: warmup is Oct–Dec 2024, train Jan–Oct 2025, validation Nov–Dec 2025, holdout Jan–Feb 2026.
- **Customer overlap:** 2,429 of 2,480 customers appear in more than one split. That's normal for a time-based split, but customer-level features or target encoding will leak across splits. 50 customers appear only in holdout (cold start). Important while building the model.
- **Volume roughly doubles** from about 3.3k transactions a month in Oct–Dec 2024 to about 7.2k a month from Jan 2025.
- Timestamps are in UTC. The "18:xx" boundaries are midnight in IST, and the dip from 18:00 to 23:00 UTC is night-time in India. Use IST for any hour-of-day features.

4.6 Structural gaps

- **There's no label column** (no is_fraud or similar), even though the features look built for fraud detection. If the labels live in another table, it needs a join key, which is presumably transaction_id.
- 16 customers have only one transaction and 49 have fewer than five, so their velocity features are thin.

5. AI/ML opportunities: potential use cases and relevant attributes

As we are in the early stage of the discussion below is what we can follow from the project document

1. **Base model selection.** Shortlist at least three open-weight models with a commercial-use licence in the 1B–8B parameter range (for example Llama 3.x, Qwen 2.5, Mistral, Gemma). Pick one on validation PR-AUC after a small pilot fine-tune, licence terms and inference cost.
2. **Continued pre-training (domain adaptation).** Train on the ~135 M-token fraud text corpus (Section 7.3) so the model learns payments vocabulary and investigator reasoning.
3. **Supervised fine-tuning (SFT).** Train on serialised transactions with two targets: the fraud label (through a classification head on the final hidden state) and, for the 8,000 gold cases, a short JSON explanation. Use parameter-efficient methods (LoRA / QLoRA) unless full fine-tuning is shown to be clearly better.

6. Discussion notes: key observations, questions, decisions and learnings

Key Observations

1. device_id missing with UPI type transaction can be directly classified as a fraud, as it could be done through a theft device. This needs be investigated.
2. If the device location permission is off then whether the app can handle the transaction without location permission? And is it needed and customers privacy also matter sometimes
3. **Customer and merchant cities barely match:** only 6% are the same city, and that holds even for grocery and quick-commerce (6%), where you'd expect local merchants.

Questions

A. We want understand more about is_syenthic field in the different tables.

B. A payer far from the merchant, or in a new city, raises risk - Why is this a risk. We need more clarifications (refer 3rd Key observation)

C. Night transactions are not always risky. This needs to be clarified further and we need other points as well.

D. Why is average only calculated for 90 days

7. Individual contributions: what each pod member explored and explained

Both of us (Naga Teja and Krishna) worked together throughout the doc and discussed each area in detail. Hence we explored the complete data together and explained each point in detail. We worked as a team.

Naga Teja:

Krishna